

And $[x_i, h_i] \rightarrow [w_i, V] \rightarrow \text{value vector}$

* Multi-head Transformer

- Problem with self attention

→ Eg. The man saw the astronomer with a telescope

This sentence has two different meanings:-

- 1) A man through telescope saw an astronomer
- 2) A man saw an astronomer holding a telescope

- Self attention tries to capture single type of meanings, most probably 1st meaning.

- But we also want our algorithms to interpret all types of meaning a sentence may have.

- Hence, multi-head attention came in picture.

- Multi-head attention uses self attention and its weights for more than one time.

- i.e. First it uses different set of query, key and value weights and trains it completely, then during 2nd time training it uses another set of random query, key and value weights and then trains it again completely from scratch, the final vector could generate another meaning of said sentence and so on.

